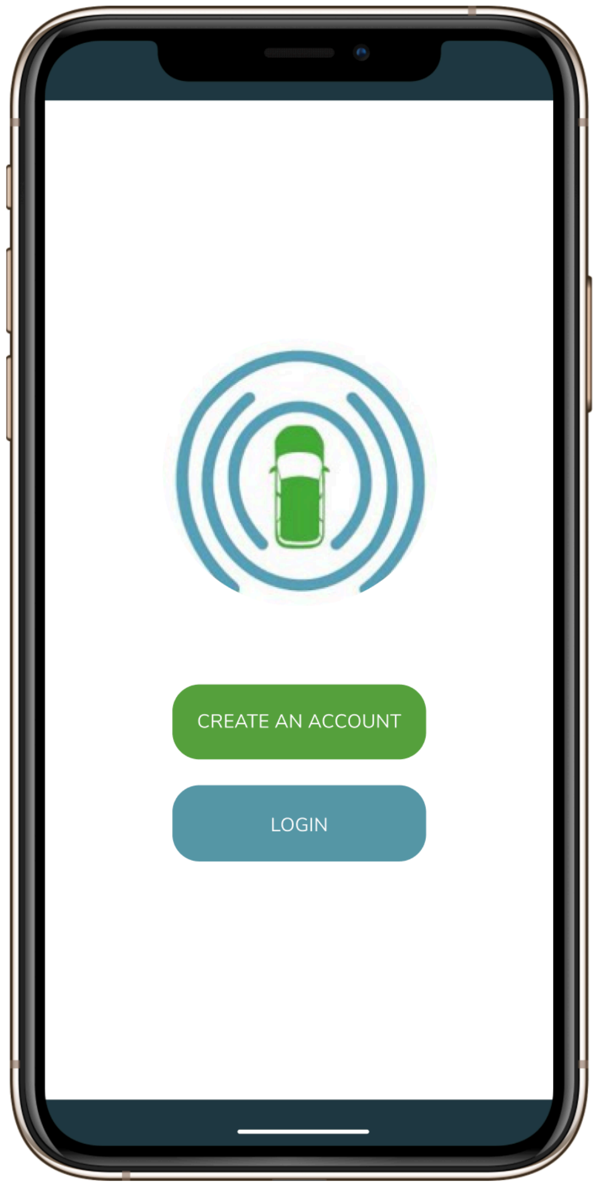


Designing an Onboarding Experience for a Semi-Autonomous Car-Sharing Vehicle

A UX research and product design project exploring how renters can safely learn unfamiliar vehicle features before driving.

As vehicles become more connected and automated, the challenge is not just designing the technology - it is helping people understand, trust, and use that technology safely.

Through the University of Washington EcoCAR team, I worked on the HMI/UX team for the EcoCAR Mobility Challenge, a four-year competition sponsored by General Motors. The competition challenged university teams to retrofit a 2019 Chevrolet Blazer into an electric-hybrid, semi-autonomous vehicle intended for a car-sharing platform. My team focused on the user-facing mobile app experience, specifically how renters could learn about unfamiliar semi-autonomous features before getting behind the wheel.

Project Snapshot

Project / Client: University of Washington EcoCAR Mobility Challenge
Timeline: Year 3, September 2020 – June 2021
Role: Design Team Lead, UX/UI Researcher & Designer
Team / Stakeholders: HMI/UX Design Team, App Development Team, broader UW EcoCAR team, General Motors competition stakeholders
Problem Space: Helping car-sharing users learn unfamiliar semi-autonomous vehicle features before operating a hybrid, semi-autonomous vehicle
Methods: Competitive analysis, user research, survey research, interviews, personas, information architecture, value/feasibility mapping, wireframing, low-fidelity prototyping
Tools: Figma, Google Forms, Miro
Outputs: Competitive analysis, personas, IA flows, renter-side wireframes, low-fidelity app screens, Adaptive Cruise Control onboarding flow, and Year 3 final presentation
Outcome / Impact: UW EcoCAR placed 4th out of 11 competitors in Year 3; judges highlighted the team's accessibility considerations, research process, and clear design rationale
Focus Areas: Automotive UX, human factors, onboarding, accessibility, emerging technology, safety-sensitive UX

The Challenge

Car-sharing users may rent vehicles they have never driven before. When those vehicles include semi-autonomous features, such as Adaptive Cruise Control, the learning curve becomes more than a convenience issue - it becomes a safety and trust issue.

Our design challenge was to help renters quickly and confidently understand how to use unfamiliar vehicle features without relying on a staff member, printed manual, or third-party support.

Our initial design question was:

> **How can a car-sharing app aid in educating new renters about semi-autonomous features they might not have experience with?**

As we learned more through research, we refined the question to:

> **How might users of car-sharing services achieve sufficient education on semi-autonomous vehicle features so they can safely and successfully operate an unfamiliar car-sharing vehicle?**

My Role

I led the design team responsible for the renter-side mobile app experience. My work included research synthesis, competitive analysis, ideation, information architecture, wireframing, low-fidelity prototyping, and design decision-making.

As design team lead, I also helped coordinate our work with the broader HMI/UX team and app development team. Because the project sat at the intersection of software, vehicle systems, and user education, our design decisions needed to account for usability, accessibility, safety, technical feasibility, and competition requirements.

Context: The EcoCAR Mobility Challenge

The EcoCAR Mobility Challenge is a multi-year collegiate engineering and design competition. The University of Washington team participated alongside other universities to transform a conventional vehicle into a hybrid, semi-autonomous vehicle for use in a car-sharing platform.

Within the broader team, the HMI/UX group was responsible for designing the digital systems that would help users interact with the vehicle. This included the mobile app, in-car interfaces, onboarding flows, and other touchpoints that could make the vehicle feel more intuitive and accessible to renters.

Why Onboarding Mattered

Unlike a privately owned car, a car-sharing vehicle may be used by many different people with different driving backgrounds, confidence levels, and familiarity with advanced vehicle features.

That created a central UX challenge:

How do we give users enough information to feel prepared without overwhelming them before a ride?

The onboarding experience needed to be:

Efficient enough for a car-sharing context
Accessible for different user proficiencies
Flexible across learning styles
Clear enough to support safe vehicle operation
Integrated with both the mobile app and in-car system

Competitive Analysis

We began by studying existing car-sharing apps to identify common patterns, strengths, and gaps. Our analysis looked at how apps supported search, reservations, payments, vehicle access, damage reporting, and pre-ride education.

We found that successful car-sharing experiences often included:

Clear map-based search with vehicle and landmark indicators
Flexible reservation flows for immediate or future bookings
Multiple payment options
Damage documentation before and after a ride
Clear instructions for unlocking and accessing the vehicle
QR code-based access with backup code entry
Consistent visual systems and navigation patterns

This helped us understand the baseline expectations users may bring into a car-sharing app. From there, we could identify where education around semi-autonomous features needed to fit into the broader rental flow.

User Research

To understand user needs and learning preferences, we conducted survey research and user interviews. We wanted to learn how people prefer to absorb new information, especially when learning about an unfamiliar product or system.

Our research found that participants valued:

Efficiency - they wanted information quickly and without unnecessary friction
Accessibility - they wanted instructions that were easy to understand and use

Flexibility - they wanted options depending on their context and comfort level
Variation in learning modes - they wanted information presented in more than one way

We also found that many participants identified as visual learners, followed by physical, verbal, and aural learners. This finding shaped our decision to include multiple ways of presenting onboarding information, including videos, images, GIFs, and short text explanations.

Elise Mathew

Age:
18 to 24 years old

Gender:
Female

Highest level of education:
High school

Occupation:
College student

Bio

Elise is a 20 year old undergraduate student at the University of Washington, studying Laws, Society, and Justice. She lives in a house with 7 other women that's a 15+ minute walk away from campus.

Personality

- *Friendly*
- *Outgoing*
- *Happy*
- *Tech savy*
- *Reasonable*
- *Motivated*

Motivations

- *Finding a balance between academics & self care*
- *Taking advantage of all opportunities given*
- *Getting work done in an effecient manner*
- *Being safe walking alone at night*

Frustrations

- *Walking back to the house alone at night after studying*
- *Trying to rent a bike that ends up being unavailable*
- *Wanting to get somewhere quickly but not having a car or reliable & accessible transportation*

Mary Edwards

Age:
50-60 years old

Gender:
Female

Highest level of education:
Masters

Occupation:
Consultant

Bio

Mary is a 59 year old who is married and a mother of 2 girls. She lives in a suburban house with her husband and dog in Portalnd, Oregon. She commutes to work in her electric car.

Personality

- Kind -Frugal - Not extremely tech savy

Motivations

- Finding a balance between work & home life
- Getting work done in an effecient manner
- Having time to relax
- Having time to get into yoga

Frustrations

- Public transportation is not accessible & efficient (does not come often enough by house & lengthens a commute time)
- Stress of driving in traffic or bad weather to & from work
- Finding street parking when going to an event (i.e. Thorns soccer game)

Target User

Based on our research, we focused on a user who regularly uses transportation to commute or access outdoor activities but may have limited familiarity with semi-autonomous vehicle features.

This user may be comfortable using mobile apps and car-sharing services, but still need support understanding how specific vehicle features work, especially if those features vary across manufacturers or models.

The design needed to support users who were curious and capable, but not necessarily technically familiar with systems like Adaptive Cruise Control.

Key Research Insights

1. Users wanted education to be quick and practical.

In a car-sharing context, users are not looking for a long training module. They need just enough information to feel prepared, confident, and safe before starting the ride.

This pushed us toward a brief onboarding flow rather than a dense educational experience.

2. Different users learn in different ways.

Because participants valued different learning modes, we decided the app should not rely only on text. The onboarding experience needed to support visual, physical, and verbal learning preferences through a mix of media and interaction.

3. Semi-autonomous features require clear mental models.

Features like Adaptive Cruise Control can be confusing if users do not understand what the feature does, when to use it, how to activate it, and what limitations it has. We needed to make the feature feel understandable without oversimplifying safety-critical information.

4. The app and vehicle experience needed to work together.

Because users would move between the mobile app and the physical vehicle, the app could not be designed in isolation. The experience needed to prepare users before the ride while also connecting to what they would see and do inside the car.

Design Principles

Based on our research, we created a set of principles to guide the onboarding and app experience.

Make learning efficient.

Users should be able to understand key vehicle features quickly, especially when they are preparing for a reservation.

Support multiple learning styles.

The experience should use a combination of text, visuals, videos, GIFs, and review moments so users can learn in the format that works best for them.

Prioritize safety and confidence.

The onboarding flow should help users understand not only how to turn features on and off, but also when and how to use them appropriately.

Reduce reliance on outside support.

Users should be able to complete onboarding independently without needing a staff member, manual, or third-party explanation.

Design for a range of proficiencies.

The experience should work for users who are comfortable with advanced vehicle technology and users who are encountering semi-autonomous features for the first time.

Connect app-based learning to in-car behavior.

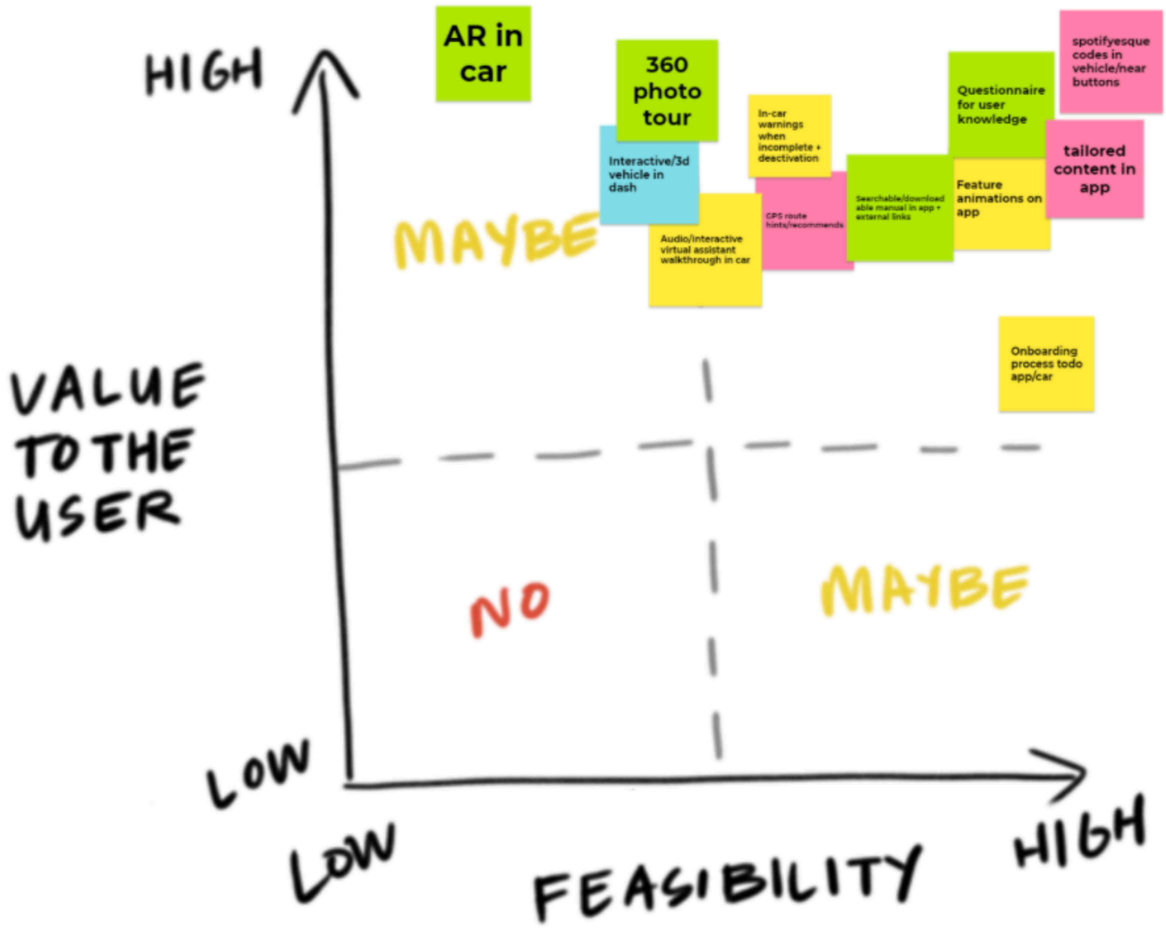
The app should prepare users for what they will physically see and do inside the vehicle.

Ideation

After synthesizing research, we brainstormed possible onboarding experiences and mapped them by user value and feasibility.

We considered how much information users needed, when they should receive it, and what formats would best support safe learning. From this process, we decided to include a brief onboarding experience that used a combination of videos, images, GIFs, and text to explain vehicle features.

For the competition, we were required to onboard users on Adaptive Cruise Control. We also saw an opportunity to design the system so it could support other vehicle features, such as windshield wipers, lights, or controls that may vary across manufacturers and models.



Caption: *We mapped onboarding ideas by value and feasibility to identify which concepts could best support safe, efficient learning.*

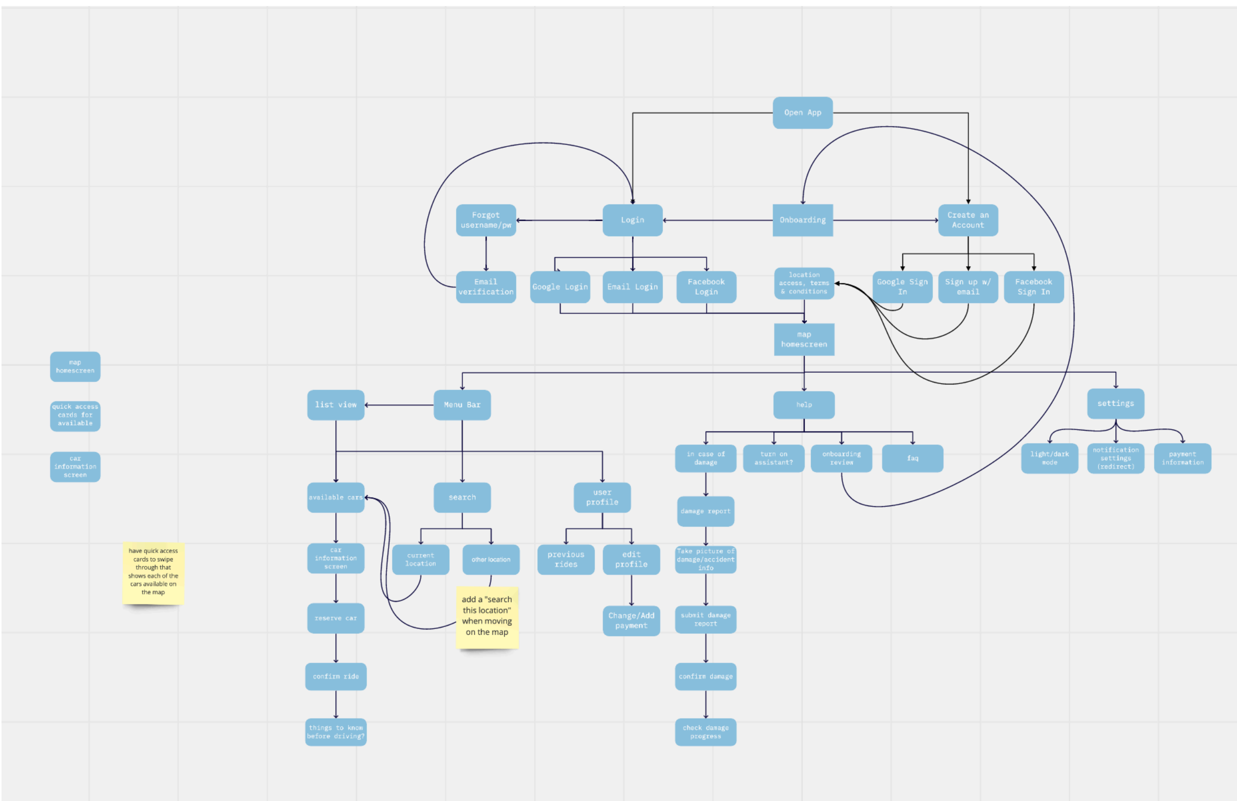
Information Architecture

We created information architecture flows for both the renter side and owner side of the app. This helped us understand how different user types would move through the product and where onboarding needed to fit within the broader car-sharing experience.

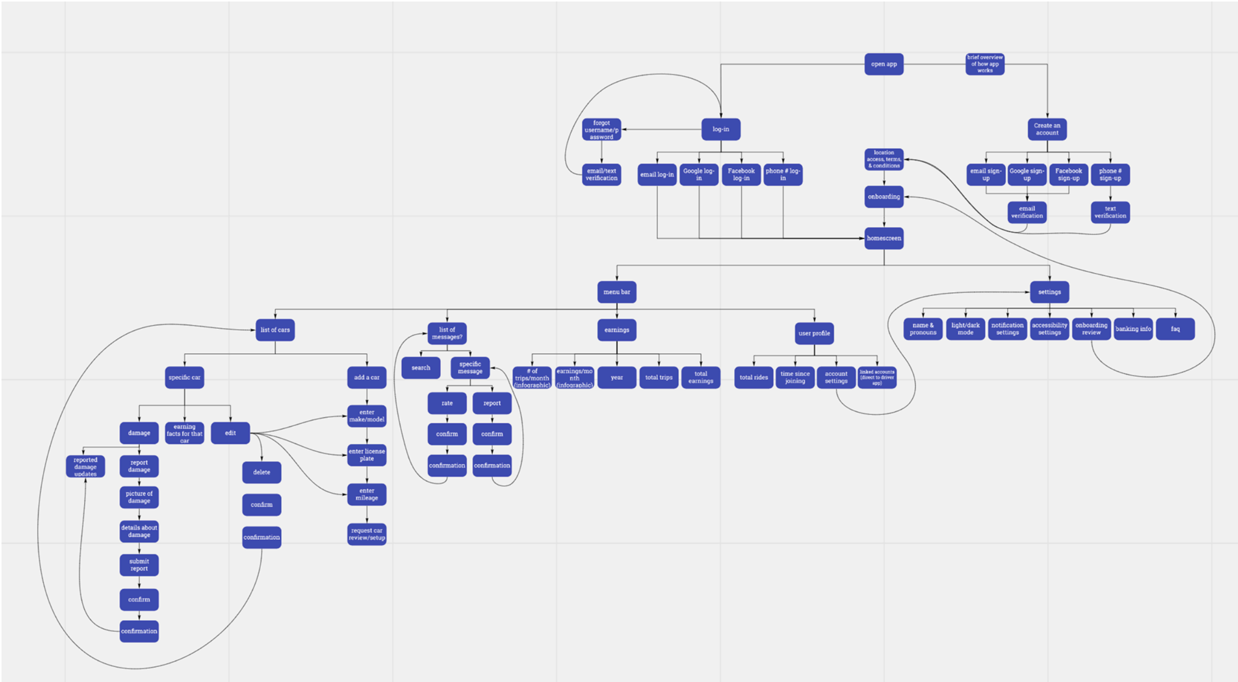
One challenge we identified was that connecting renter-side and owner-side flows could create unnecessary complexity. To simplify the experience, we separated the two flows while still allowing users to switch between them within the app.

For the renter experience, the app needed to support:

Finding a vehicle
Booking a reservation
Accessing vehicle information
Completing onboarding
Learning semi-autonomous features
Unlocking the vehicle
Reviewing feature instructions when needed



Owner Side Flow



User Side Flow

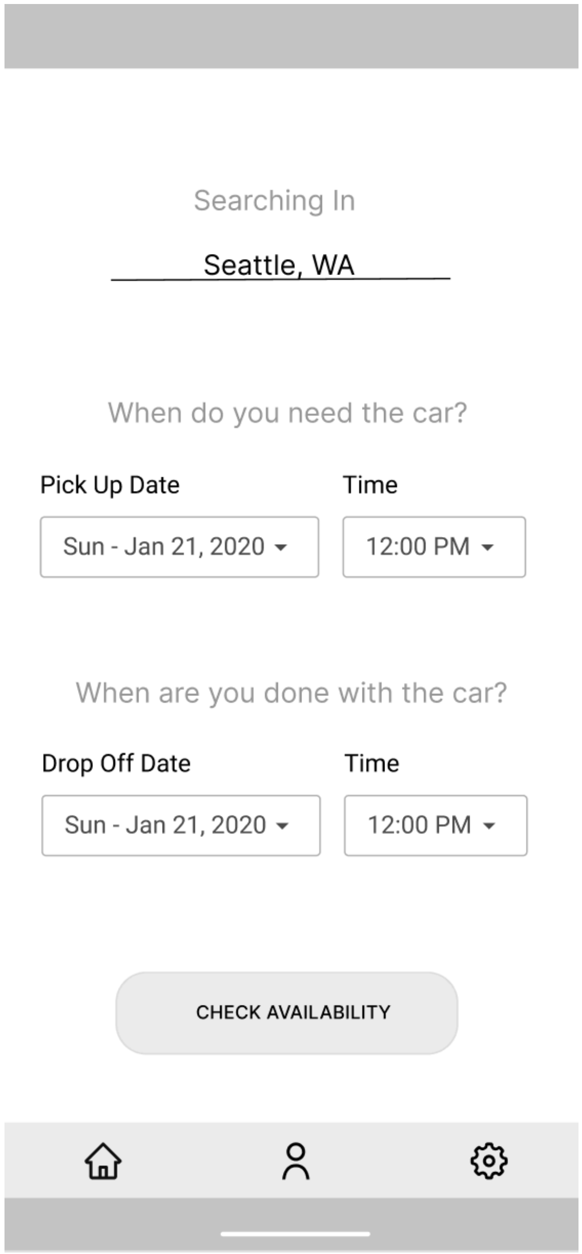
Caption: *The IA helped us separate renter and owner flows while identifying where education should appear in the car-sharing journey.*

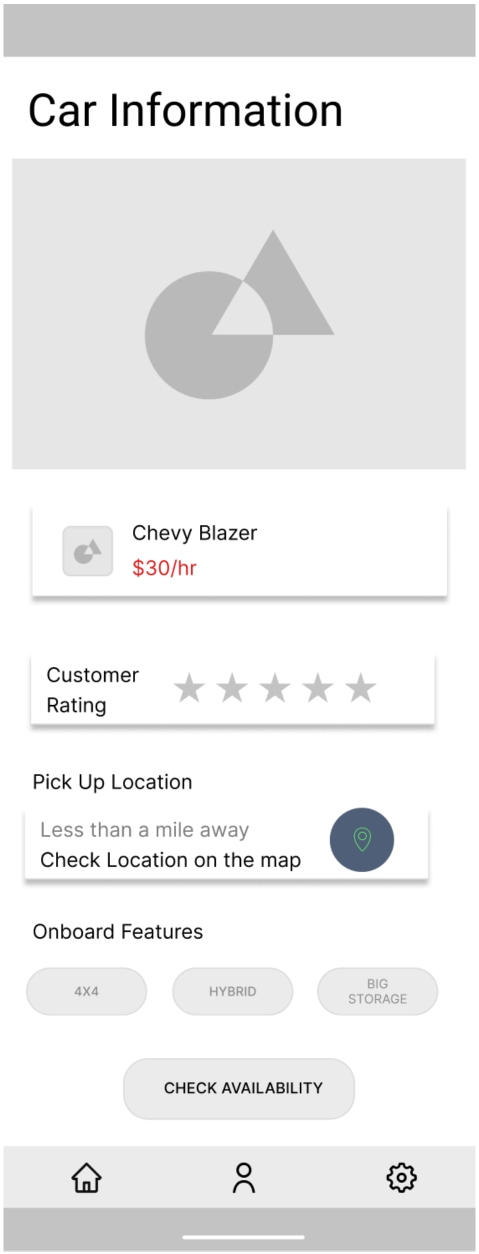
Wireframing

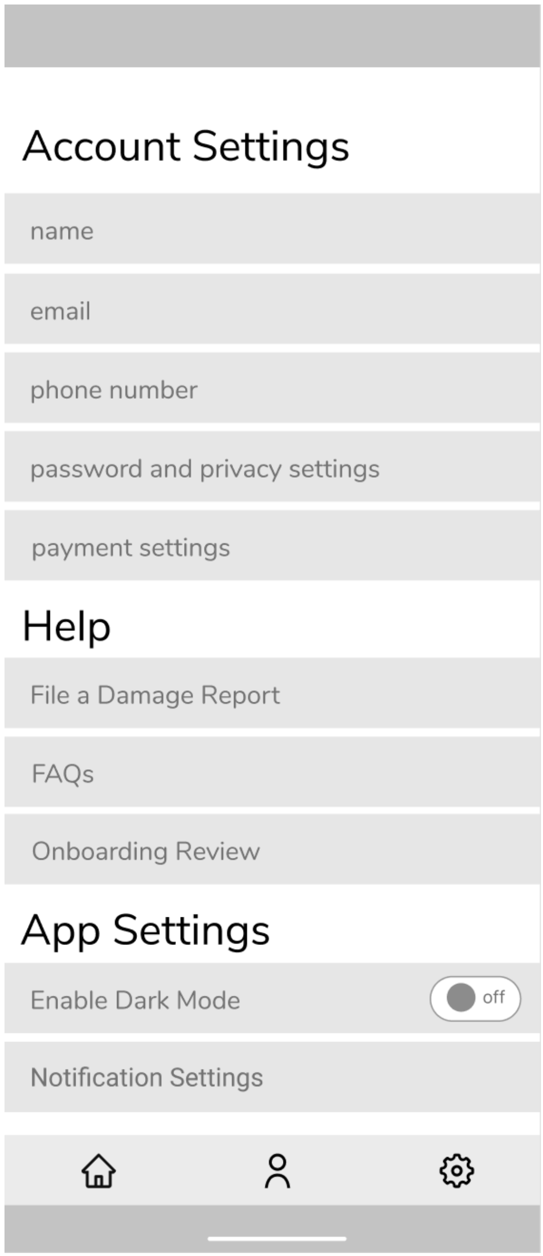
We created wireframes for the renter-side experience, focusing on making the interface feel seamless, accessible, and informative.

The wireframes helped us explore how users would move from booking a car to learning about its features. They also gave us a way to evaluate whether important information was visible at the right moments in the journey.

I personally designed the Settings screen (please see below), using familiar patterns from popular mobile apps so the experience would feel intuitive and easy to navigate.


Searching In
Seattle, WA
When do you need the car?
Pick Up Date
Time
Sun - Jan 21, 2020
12:00 PM
When are you done with the car?
Drop Off Date
Time
Sun - Jan 21, 2020
12:00 PM
CHECK AVAILABILITY


Car Information
Chevy Blazer
$30/hr
Customer Rating
Pick Up Location
Less than a mile away
Check Location on the map
Onboard Features
4X4
HYBRID
BIG STORAGE
CHECK AVAILABILITY

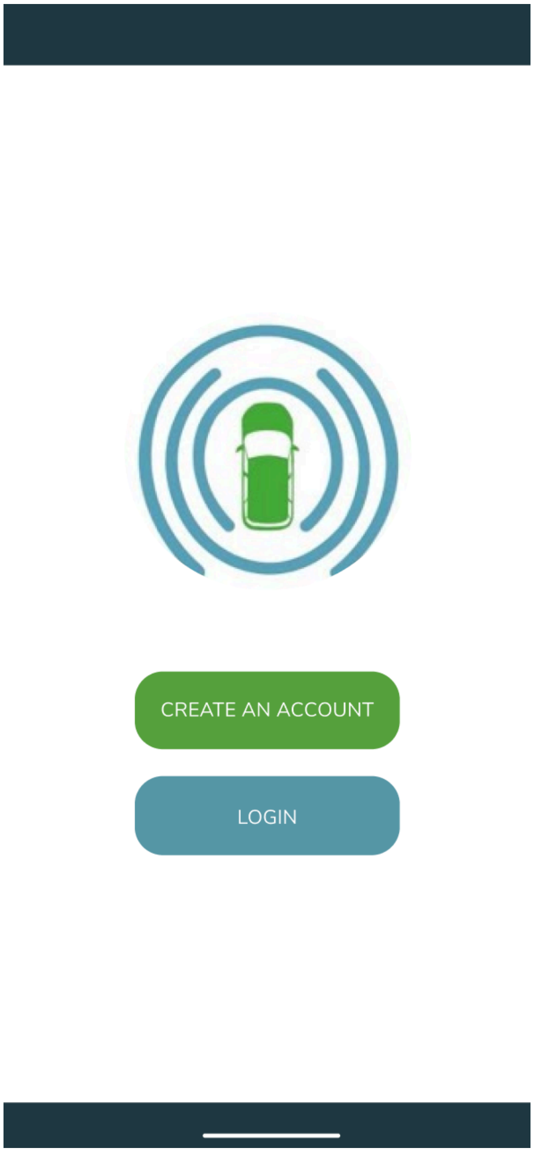


Caption: *Wireframes helped us define the renter-side experience before moving into more detailed visual design.*

Low-Fidelity Prototyping

We then moved into low-fidelity prototypes for the renter-side app experience. These screens incorporated early branding decisions, including color palette, typography, and icons.

I designed several renter-side screens, including examples shown in the original project, with input from the broader team. The goal was to create an interface that felt accessible, informative, and easy to move through before or during a rental experience.



Chevrolet, Blazer

1A9-Q23E

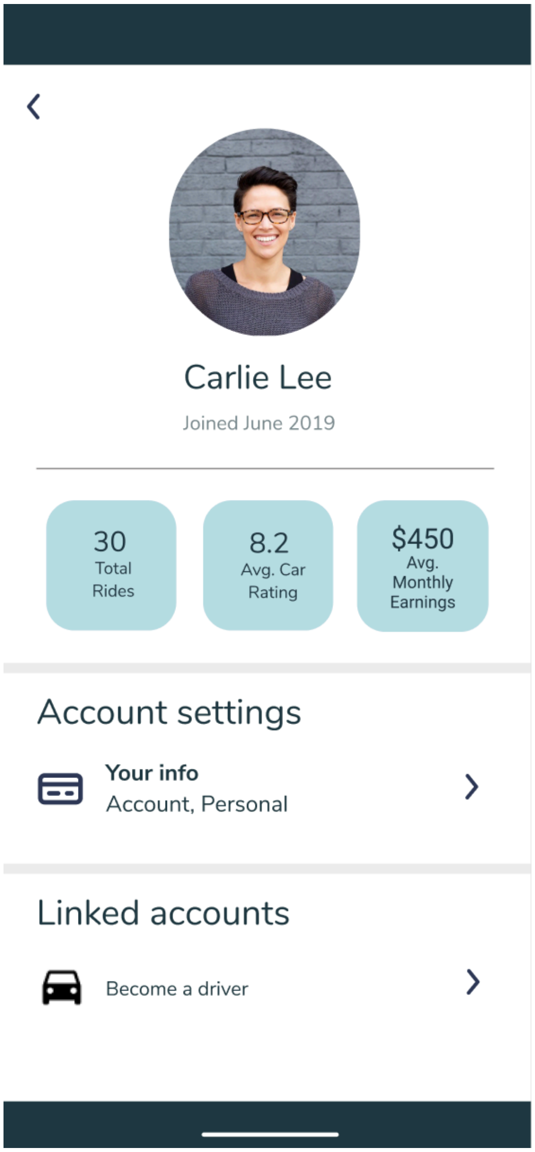
7.5/10 10 reviews

$45/day

Availability

January

S	M	T	W	Th	F	Sa
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1



Caption: *The low-fidelity prototype helped us explore how renters would navigate the app and access vehicle information.*

Onboarding Flow

The onboarding flow was the core of our solution.

For Year 3, our team focused on onboarding users to Adaptive Cruise Control. The flow introduced the feature, explained its purpose, and gave users a way to review how it worked before operating the vehicle.

I designed several onboarding screens, including the start screen, overview screen, feature review screen, and progress indicator (please see below). The flow was designed to break down technical information into smaller, more approachable steps so users could build understanding without feeling overwhelmed.

Overview

- Adaptive Cruise Control (ACC)
- Windshield Wipers
- Parking Break
- Trunk
- Gas Cap

Start

Caption: *The onboarding flow used short, step-based education to help renters understand Adaptive Cruise Control before driving.*

Final Concept

The final app concept helped renters move through the car-sharing journey while receiving the education needed to safely operate an unfamiliar semi-autonomous vehicle.

The experience included:

Vehicle search and reservation
Renter-side account and settings flows
Vehicle information
Pre-ride onboarding
Adaptive Cruise Control education
Multiple content formats for different learning styles
A structure that could expand to other vehicle features

Rather than treating onboarding as a separate training module, we designed it as part of the rental experience - something users could complete before driving and revisit when needed.

Why This Experience Was Different

Many car-sharing apps focus on access: finding a car, booking it, unlocking it, and ending the ride.

Our concept extended that experience to include **feature education**. This was especially important because the vehicle was not just unfamiliar - it included semi-autonomous capabilities that users might not have encountered before.

The app was designed to help renters answer practical questions before driving:

What does this feature do?
When should I use it?
How do I turn it on or off?
What should I expect once I'm inside the vehicle?
Where can I review the information if I forget?

That made the product less about passive instruction and more about supporting user confidence in a real-world, safety-sensitive context.

Competition Outcome

At the end of Year 3, the University of Washington EcoCAR team placed **4th out of 11 competitors**.

Judges responded positively to the team's design process, accessibility considerations, and explanation of design decisions. They specifically noted the team's thoughtful approach to accessibility, use of remote research methods during COVID-era constraints, and clear progression from research findings to prototype iterations.

This feedback reinforced the value of grounding the app experience in user needs, learning preferences, and clear design rationale.

Judge 1:
"LOVED the accessibility considerations. Above and beyond!
The extra comment about the word "onboarding" was refreshingly honest.
For those unfamiliar with human factors, psychology, or user-centered design, this swim-lane can pose some serious challenges. That did not come across at all! The team is really understanding the concepts and is on the right path, keep up the good work, I look forward to seeing the progress you make over the next year :)"

Judge 2:
"Good: Thorough reasoning provided throughout the presentation.
Appreciated the concept to focus on real world interactions. Excellent job with the Think Aloud technique and purpose for checking for understanding."

Judge 3:
"I like how you designed alternative presentation methods for different learning styles, and explained WHY you did that. Nice use of tools for remote user research given COVID challenges.
Good clear examples of what users had trouble with; demonstrated that you learned and the 1st prototype was useful.
Nice progression from user research, to a conclusion about how to attack prototype scope, and iteration between prototype versions."

View Year 3 final presentation.

Reflection

This project helped me understand how UX research and design can support safety, trust, and education in complex technical systems.

Unlike a typical consumer app, this experience needed to account for the physical context of use. A renter would not just be tapping through screens - they would eventually be operating a real vehicle with unfamiliar semi-autonomous features. That made clarity, accessibility, and timing especially important.

As design team lead, I also learned how to guide a team through a highly cross-functional project. Our work needed to align with competition requirements, engineering constraints, HMI considerations, and user needs. That experience helped me become more comfortable translating research into practical design decisions within a larger technical ecosystem.

Looking back, this project strengthened my interest in designing experiences that help people understand and confidently use complex products, especially when the stakes are higher than simple convenience.

What I'd Do Differently Today

If I were approaching this project now, I would spend more time testing the onboarding flow with users in realistic scenarios.

For example, I would want to evaluate:

Whether users understood Adaptive Cruise Control after completing the onboarding flow
Which content formats best supported comprehension and recall
Whether users could connect app instructions to controls inside the vehicle
How much information users needed before driving versus during the ride
Where safety-critical information should be reinforced

I would also want to test comprehension more explicitly, perhaps through scenario-based tasks or knowledge checks, to understand whether the onboarding experience actually prepared users to use the feature safely.

Even with those limitations, this project gave me valuable experience designing for emerging technology, accessibility, onboarding, and the relationship between digital interfaces and real-world behavior.

Short Version for the Project Card

Designing Onboarding for a Semi-Autonomous Car-Sharing Vehicle
Led UX research and design for the renter-side mobile app experience on UW's EcoCAR team. Designed onboarding flows to help car-sharing users understand unfamiliar semi-autonomous

vehicle features, including Adaptive Cruise Control. The team placed 4th out of 11 competitors, with judges noting the project's accessibility considerations and clear research-to-design rationale.

Tags

UX Research · Product Design · Human Factors · Automotive UX · Onboarding · Accessibility · Competitive Analysis · Information Architecture · Wireframing · Prototyping · Emerging Technology · Car Sharing · Safety-Critical UX